TED (15) 5061 **model question paper** Reg. No……………………………

(Revision 2015) Signature………………………….

FIFTH SEMESTER DIPLOMA EXAMINATION IN TEXTILE TECHNOLOGY

FABRIC FORMATION - III

[Time : 3hrs]

(Maximum marks : 100)

PART - A

(Maximum marks : 10) Marks

I Answer the following questions in one or two sentences. Each question carries two marks.

1. List the classes of box motion.
2. Define weft bunch.
3. State two features of shuttle less weaving.
4. Mention the function of throttle valve in water jet picking.
5. Define efficiency of loom

(5 x 2 =10)

PART – B

(Maximum marks: 30)

II. Answer any five of the following questions. Each questions carry 6 marks

1. Mention the principles of terry motion.
2. State the features of automatic loom
3. Compare conventional method of weaving and shuttle less weaving.
4. List the classification of rapier loom.
5. Describe the functional parts of air nozzle.
6. Discuss the advantages and disadvantages of water jet loom.
7. Calculate the length of cloth that can be produced from 1Kg cone of 30^s cotton yarn Reed width 48"and PPI is 40.

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(5 x 6 = 30)

PART – C

(Maximum marks : 60)

(Answer one full question from each unit. Each full question carries 15 marks.)

UNIT I

III a) Describe the working of a circular box motion with the help of sketch. (12)

b) State the use of thread cutters in automatic loom. (3)

OR

IV a) Explain the working of shuttle protector mechanism with the help of figure. (12)
b) Mention the classes of terry motion. (3)

UNIT II

V. a) Draw and describe the working of torsion bar picking mechanism. (12)
b) Write the classification of shuttle less loom. (3)

OR

VI. a) Describe the basic requirements of shuttle less weaving including warp and weft preparation. (12)
b) Discuss the quality of Sulzer woven cloth. (3)

UNIT III

VII. a) Describe the passage of yarn through Maxbo air jet loom with sketch. (10)
b) Describe the features of water nozzle. (5)

OR

VIII. a) With the help of sketches describe the stages of weft insertion in waterjet loom. (10)
b) List the requirements for air jet weft insertion. (5)

UNIT IV

IX. a) Calculate the total weight of yarn in a piece of cloth which is 50" wide in the reed. The cloth length is 50 yards and the ends/ inch and picks/inch are 56 and40 respectively. The count of warp is 30^s and weft is 36^s. Selvedge ¼" on each side drawn 4ends/dent and body ends drawn 2ends/dent. The warp regain is 4 %. (15)

OR

X. a) Find the time required to exhaust weavers beam from the given data
Length of warp on the beam - 1200 yards   warp regain – 4%   waste – 5 yards
Loom speed -190 ppm   PPI – 40   Efficiency – 75 % (10)
b) The cloth produced in a loom/ hour is 6 yards. If the speed of loom is 210rpm and PPI in the cloth is 44, find efficiency of loom. (5)

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